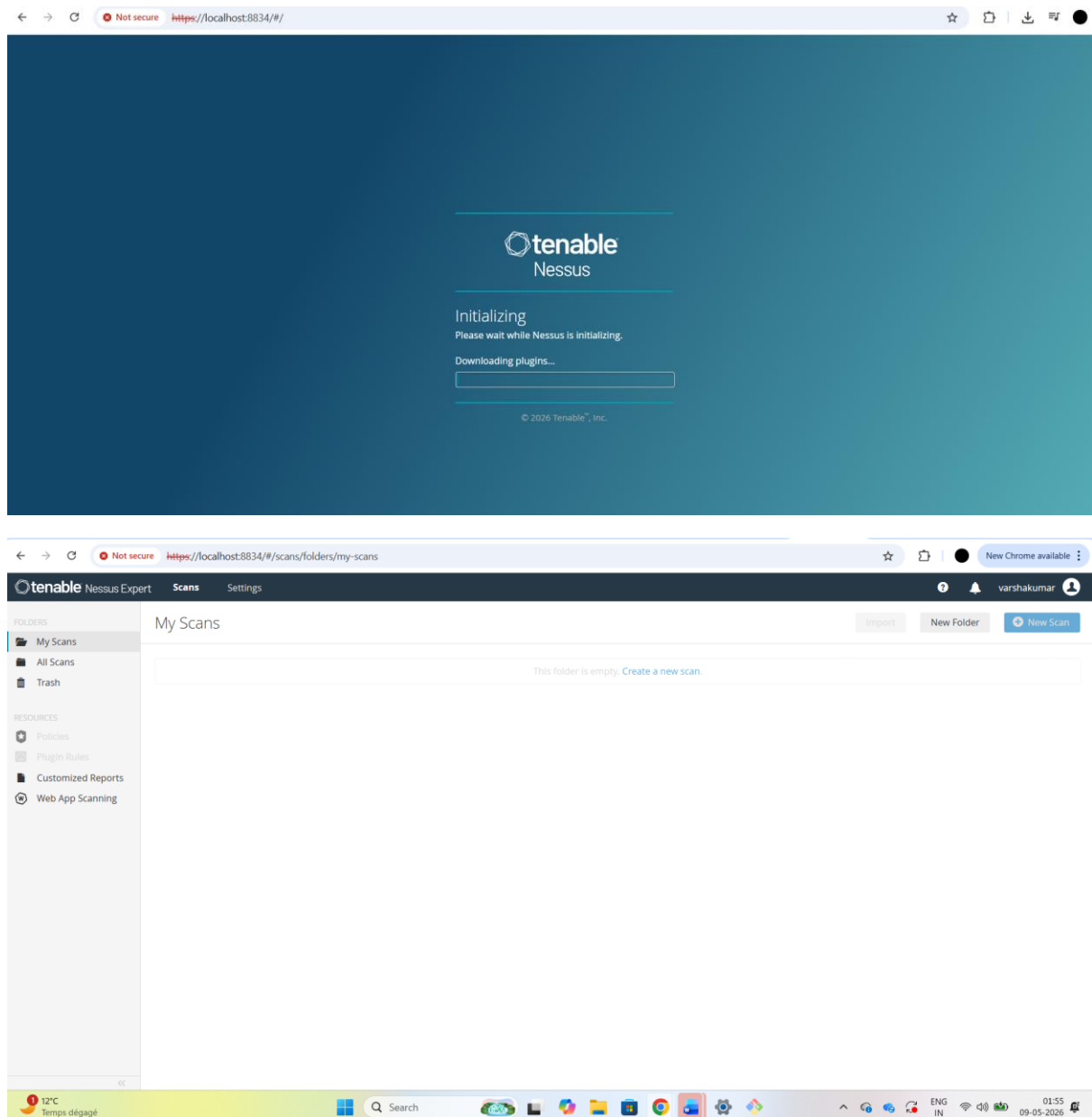


Perform a Basic Vulnerability Scan on Your PC.

Step 1:

Intallization of Nessus



Step 2:

Set the Scan Target

```

kk641@LAPTOP-1A73Q813 MINGW64 ~
$ ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::e107:8ce6:234:6263%7
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a864:c529:9dd:31bb%22
    IPv4 Address. . . . . : 192.168.70.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::6bea:b224:1bc7:5e28%20
    IPv4 Address. . . . . : 192.168.74.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2a01:e0a:1041:70e0:b3aa:2195:2c86:d503
    Temporary IPv6 Address. . . . . : 2a01:e0a:1041:70e0:1ce4:d059:3ffe:9699
    Link-local IPv6 Address . . . . . : fe80::1bf8:4b80:25f8:96bb%14
    IPv4 Address. . . . . : 192.168.0.45
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::f6ca:e5ff:fe41:4cb7%14
                                192.168.0.254

Ethernet adapter Bluetooth Network Connection:

```

IPv4 Address. : 192.168.70.1

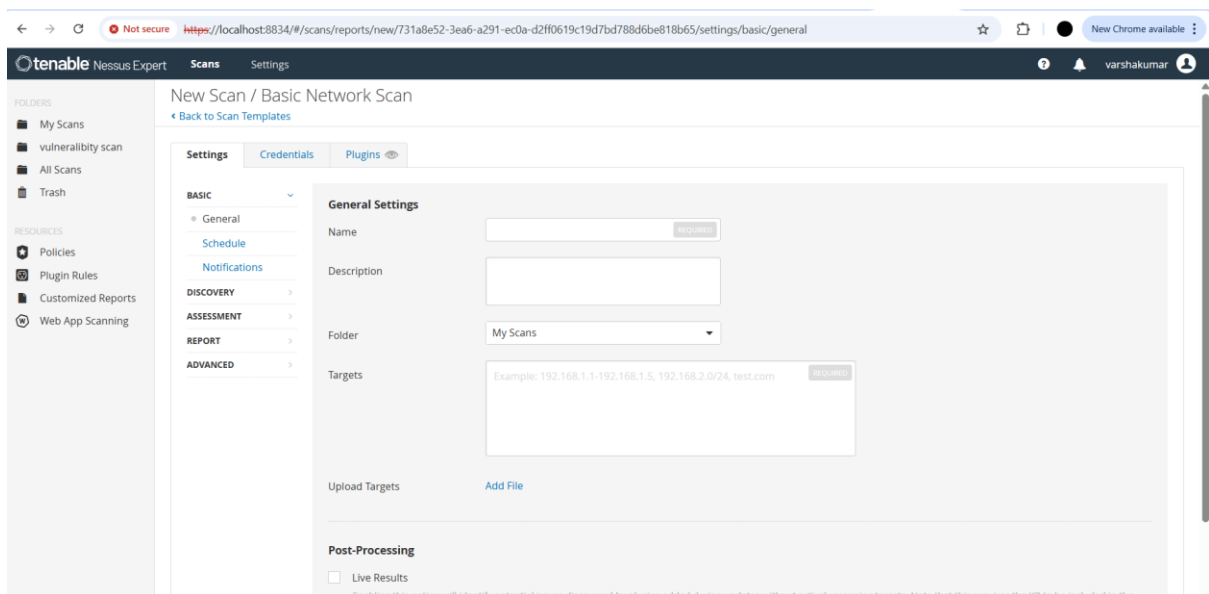
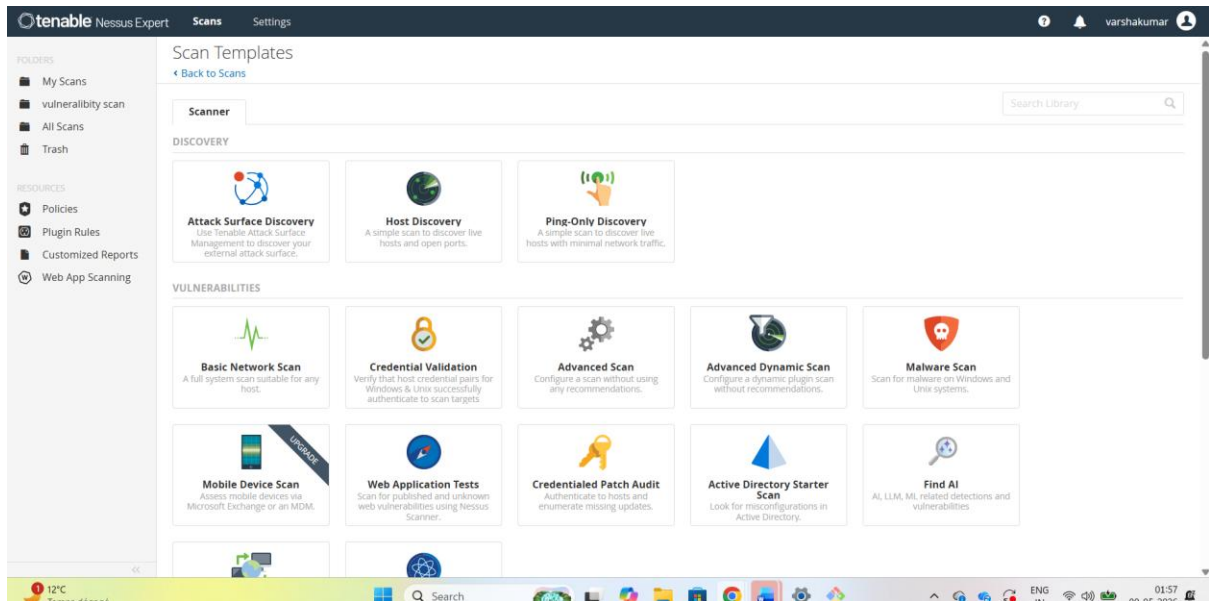
Step 3: Start a Full Vulnerability Scan

During the scan:

The tool checks:

- Open ports
- Weak services
- Missing updates
- Security issues

Click on basic network scan



Find the ip address and Enter the Target ip and select the basic network scan – save

The screenshot shows the 'New Scan / Basic Network Scan' configuration page in Tenable Nessus Expert. The left sidebar contains 'FOLDERS' (My Scans, vulnerability scan, All Scans, Trash) and 'RESOURCES' (Policies, Plugin Rules, Customized Reports, Web App Scanning). The main content area has tabs for 'Settings', 'Credentials', and 'Plugins'. Under 'Settings', there are sections for 'BASIC' (General, Schedule, Notifications), 'DISCOVERY', 'ASSESSMENT', 'REPORT', and 'ADVANCED'. The 'General Settings' section is active, showing fields for 'Name' (My PC Scan), 'Description', 'Folder' (My Scans), and 'Targets' (192.168.70.1). There is an 'Upload Targets' section with an 'Add File' link. At the bottom, the 'Post-Processing' section has a 'Live Results' checkbox.

Click on launch

The screenshot shows the 'My Scans' page in Tenable Nessus Expert. The left sidebar is the same as the previous screenshot. The main content area shows a table with 1 scan. The table has columns: Name, Scan Type, Schedule, Last Scanned, and a 'Launch' button. The scan 'My PC Scan' is listed with a 'Vulnerability' scan type, 'On Demand' schedule, and 'N/A' last scanned time.

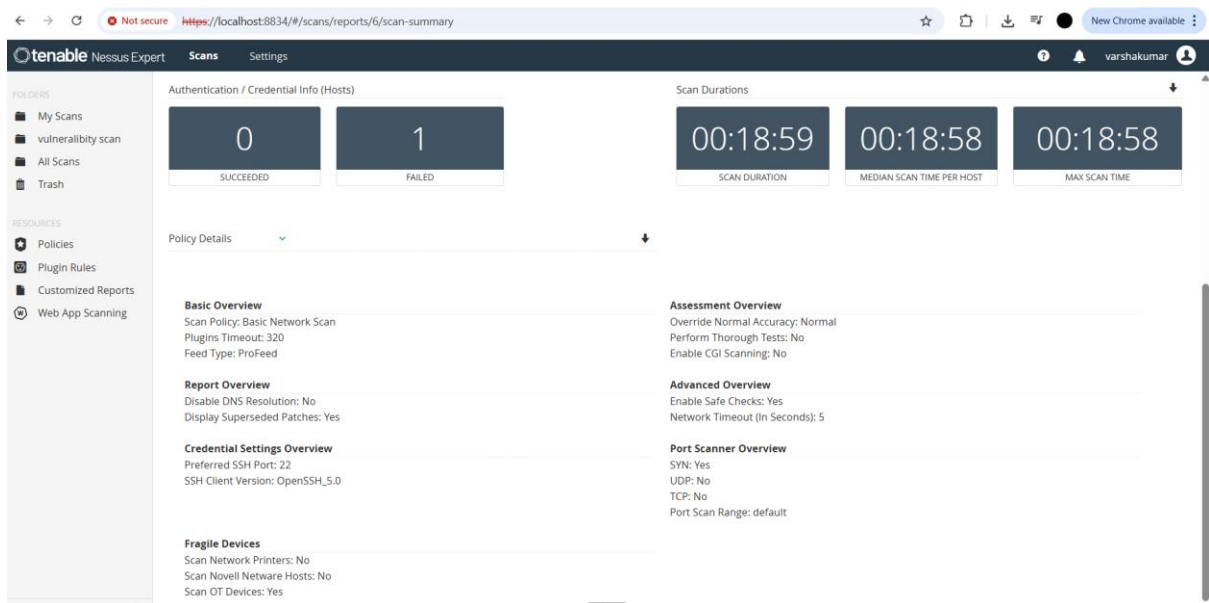
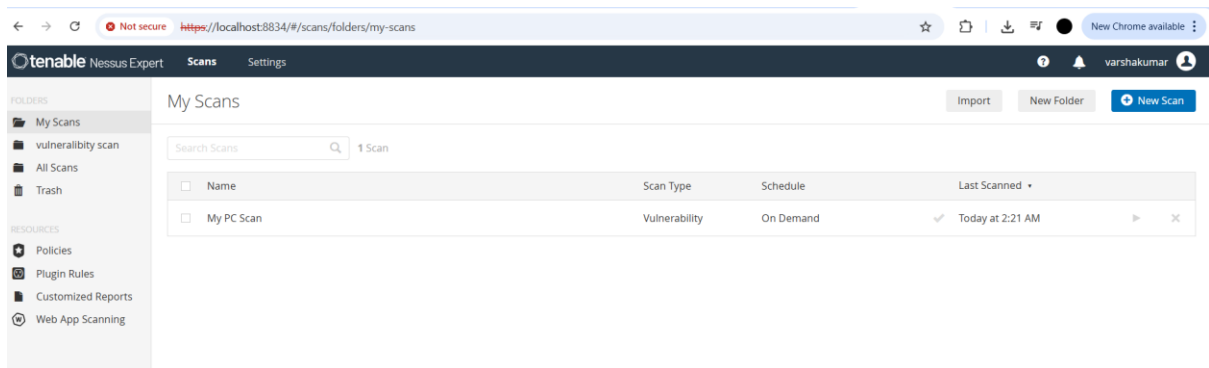
Name	Scan Type	Schedule	Last Scanned	Launch
My PC Scan	Vulnerability	On Demand	N/A	Launch

Started the scan

The screenshot shows the 'My Scans' page in Tenable Nessus Expert. The left sidebar is the same as the previous screenshot. The main content area shows a table with 1 scan. The scan 'My PC Scan' is listed with a 'Vulnerability' scan type, 'On Demand' schedule, and 'Today at 2:02 AM' last scanned time. The 'Launch' button is now a green circle with a checkmark, indicating the scan is completed.

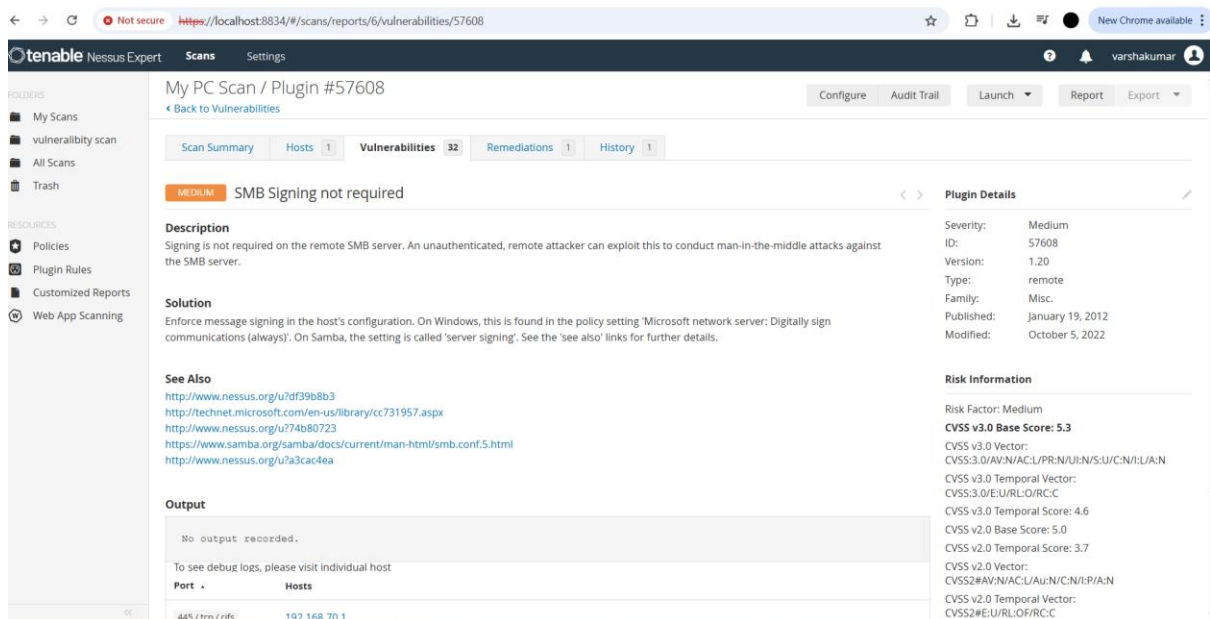
Name	Scan Type	Schedule	Last Scanned	Launch
My PC Scan	Vulnerability	On Demand	Today at 2:02 AM	Completed

And the scan is completed in 18 minutes



Step 5: Review the Results

Here we can see the description, risk and mitigation for the vulnerability.



Step 6: mitigation

“**SMB Signing not required**” is considered a **security vulnerability/misconfiguration**, especially in enterprise or networked environments.

It means your system allows SMB communication without requiring digital signing, which can make it vulnerable to:

- **Man-in-the-Middle (MITM) attacks**
- SMB relay attacks
- Packet tampering

What is SMB?

SMB = **Server Message Block**

It's the protocol Windows uses for:

- File sharing
- Printer sharing
- Network communication

Usually runs on:

Port 445

Why “SMB Signing Not Required” is risky

Without signing:

- Attackers on the same network can intercept SMB traffic
- Modify data in transit
- Relay authentication requests

Fix Applied

Enabled:

“Microsoft network server: Digitally sign communications (always)”

“Microsoft network client: Digitally sign communications (always)”

Restarted the system after applying changes.

Conclusion

The vulnerability assessment identified SMB signing configuration as a security weakness. Enabling SMB signing improves the integrity and security of SMB communications and helps protect the system against network-based attacks.